
C.2 Probability

Probability is an essential tool for the design and analysis of probabilistic and randomized algorithms. This section reviews basic probability theory.

We define probability in terms of a *sample space* S , which is a set whose elements are called *outcomes* or *elementary events*. Think of each outcome as a possible result of an experiment. For the experiment of flipping two distinguishable coins, with each individual flip resulting in a head (H) or a tail (T), you can view the sample space S as consisting of the set of all possible 2-strings over $\{H, T\}$:

$$S = \{HH, HT, TH, TT\}.$$

An *event* is a subset¹ of the sample space S . For example, in the experiment of flipping two coins, the event of obtaining one head and one tail is $\{HT, TH\}$. The event S is called the *certain event*, and the event $\emptyset$ is called the *null event*. We say that two events A and B are *mutually exclusive* if $A \cap B = \emptyset$. An outcome s also defines the event $\{s\}$, which we sometimes write as just s . By definition, all outcomes are mutually exclusive.

Axioms of probability

A *probability distribution* $\Pr \{ \}$ on a sample space S is a mapping from events of S to real numbers satisfying the following *probability axioms*:

1. $\Pr \{A\} \geq 0$ for any event A .
2. $\Pr \{S\} = 1$.
3. $\Pr \{A \cup B\} = \Pr \{A\} + \Pr \{B\}$ for any two mutually exclusive events A and B . More generally, for any sequence of events $A_1, A_2, \dots$ (finite or countably infinite) that are pairwise mutually exclusive,

$$\Pr \left\{ \bigcup_i A_i \right\} = \sum_i \Pr \{A_i\} .$$

We call $\Pr \{A\}$ the *probability* of the event A . Axiom 2 is simply a normalization requirement: there is really nothing fundamental about choosing 1 as the probability of the certain event, except that it is natural and convenient.

Several results follow immediately from these axioms and basic set theory (see Section B.1). The null event $\emptyset$ has probability $\Pr \{\emptyset\} = 0$. If $A \subseteq B$, then $\Pr \{A\} \leq \Pr \{B\}$. Using $\bar{A}$ to denote the event $S - A$ (the *complement* of A), we have $\Pr \{\bar{A}\} = 1 - \Pr \{A\}$. For any two events A and B ,

$$\Pr \{A \cup B\} = \Pr \{A\} + \Pr \{B\} - \Pr \{A \cap B\} \quad (\text{C.14})$$

$$\leq \Pr \{A\} + \Pr \{B\} . \quad (\text{C.15})$$

In our coin-flipping example, suppose that each of the four outcomes has probability $1/4$. Then the probability of getting at least one head is

$$\begin{aligned} \Pr \{HH, HT, TH\} &= \Pr \{HH\} + \Pr \{HT\} + \Pr \{TH\} \\ &= 3/4. \end{aligned}$$

Another way to obtain the same result is to observe that since the probability of getting strictly less than one head is $\Pr \{TT\} = 1/4$, the probability of getting at least one head is $1 - 1/4 = 3/4$.

Discrete probability distributions

A probability distribution is *discrete* if it is defined over a finite or countably infinite sample space. Let S be the sample space. Then for any event A ,

$$\Pr \{A\} = \sum_{s \in A} \Pr \{s\} ,$$

since outcomes, specifically those in A , are mutually exclusive. If S is finite and every outcome $s \in S$ has probability $\Pr \{s\} = 1/|S|$, then we have the *uniform probability distribution* on S . In such a case the experiment is often described as “picking an element of S at random.”

As an example, consider the process of flipping a *fair coin*, one for which the probability of obtaining a head is the same as the probability

of obtaining a tail, that is, $1/2$. Flipping the coin n times gives the uniform probability distribution defined on the sample space $S = \{H, T\}^n$, a set of size 2^n . We can represent each outcome in S as a string of length n over $\{H, T\}$, with each string occurring with probability $1/2^n$. The event $A = \{\text{exactly } k \text{ heads and exactly } n - k \text{ tails occur}\}$ is a subset of S of size $|A| = \binom{n}{k}$, since $\binom{n}{k}$ strings of length n over $\{H, T\}$ contain exactly k H's. The probability of event A is thus $\Pr\{A\} = \binom{n}{k}/2^n$.

Continuous uniform probability distribution

The continuous uniform probability distribution is an example of a probability distribution in which not all subsets of the sample space are considered to be events. The continuous uniform probability distribution is defined over a closed interval $[a, b]$ of the reals, where $a < b$. The intuition is that each point in the interval $[a, b]$ should be “equally likely.” Because there are an uncountable number of points, however, if all points had the same finite, positive probability, axioms 2 and 3 would not be simultaneously satisfied. For this reason, we’d like to associate a probability only with *some* of the subsets of S in such a way that the axioms are satisfied for these events.

For any closed interval $[c, d]$, where $a \leq c \leq d \leq b$, the *continuous uniform probability distribution* defines the probability of the event $[c, d]$ to be

$$\Pr\{[c, d]\} = \frac{d - c}{b - a}.$$

Letting $c = d$ gives that the probability of a single point is 0. Removing the endpoints $[c, c]$ and $[d, d]$ of an interval $[c, d]$ results in the open interval (c, d) . Since $[c, d] = [c, c] \cup (c, d) \cup [d, d]$, axiom 3 gives $\Pr\{[c, d]\} = \Pr\{(c, d)\}$. Generally, the set of events for the continuous uniform probability distribution contains any subset of the sample space $[a, b]$ that can be obtained by a finite or countable union of open and closed intervals, as well as certain more complicated sets.

Conditional probability and independence

Sometimes you have some prior partial knowledge about the outcome of an experiment. For example, suppose that a friend has flipped two fair coins and has told you that at least one of the coins showed a head. What is the probability that both coins are heads? The information given eliminates the possibility of two tails. The three remaining outcomes are equally likely, and so you infer that each occurs with probability $1/3$. Since only one of these outcomes shows two heads, the answer is $1/3$.

Conditional probability formalizes the notion of having prior partial knowledge of the outcome of an experiment. The *conditional probability* of an event A given that another event B occurs is defined to be

$$\Pr \{A \mid B\} = \frac{\Pr \{A \cap B\}}{\Pr \{B\}} \quad (\text{C.16})$$

whenever $\Pr \{B\} \neq 0$. (Read “ $\Pr \{A \mid B\}$ ” as “the probability of A given B .”) The idea behind equation (C.16) is that since we are given that event B occurs, the event that A also occurs is $A \cap B$. That is, $A \cap B$ is the set of outcomes in which both A and B occur. Because the outcome is one of the elementary events in B , we normalize the probabilities of all the elementary events in B by dividing them by $\Pr \{B\}$, so that they sum to 1. The conditional probability of A given B is, therefore, the ratio of the probability of event $A \cap B$ to the probability of event B . In the example above, A is the event that both coins are heads, and B is the event that at least one coin is a head. Thus, $\Pr \{A \mid B\} = (1/4)/(3/4) = 1/3$.

Two events are *independent* if

$$\Pr \{A \cap B\} = \Pr \{A\} \Pr \{B\} , \quad (\text{C.17})$$

which is equivalent, if $\Pr \{B\} \neq 0$, to the condition

$$\Pr \{A \mid B\} = \Pr \{A\} .$$

For example, suppose that you flip two fair coins and that the outcomes are independent. Then the probability of two heads is $(1/2)(1/2) = 1/4$. Now suppose that one event is that the first coin comes up heads and the other event is that the coins come up differently. Each of these events occurs with probability $1/2$, and the probability that both events

occur is $1/4$. Thus, according to the definition of independence, the events are independent—even though you might think that both events depend on the first coin. Finally, suppose that the coins are welded together so that they both fall heads or both fall tails and that the two possibilities are equally likely. Then the probability that each coin comes up heads is $1/2$, but the probability that they both come up heads is $1/2 \neq (1/2)(1/2)$. Consequently, the event that one comes up heads and the event that the other comes up heads are not independent.

A collection $A_1, A_2, \dots, A_n$ of events is said to be *pairwise independent* if

$$\Pr \{A_i \cap A_j\} = \Pr \{A_i\} \Pr \{A_j\}$$

for all $1 \leq i < j \leq n$. We say that the events of the collection are *(mutually) independent* if every k -subset $A_{i_1}, A_{i_2}, \dots, A_{i_k}$ of the collection, where $2 \leq k \leq n$ and $1 \leq i_1 < i_2 < \dots < i_k \leq n$, satisfies

$$\Pr \{A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}\} = \Pr \{A_{i_1}\} \Pr \{A_{i_2}\} \dots \Pr \{A_{i_k}\} .$$

For example, suppose that you flip two fair coins. Let A_1 be the event that the first coin is heads, let A_2 be the event that the second coin is heads, and let A_3 be the event that the two coins are different. Then,

$$\Pr \{A_1\} = 1/2,$$

$$\Pr \{A_2\} = 1/2,$$

$$\Pr \{A_3\} = 1/2,$$

$$\Pr \{A_1 \cap A_2\} = 1/4,$$

$$\Pr \{A_1 \cap A_3\} = 1/4,$$

$$\Pr \{A_2 \cap A_3\} = 1/4,$$

$$\Pr \{A_1 \cap A_2 \cap A_3\} = 0.$$

Since for $1 \leq i < j \leq 3$, we have $\Pr \{A_i \cap A_j\} = \Pr \{A_i\} \Pr \{A_j\} = 1/4$, the events A_1, A_2 , and A_3 are pairwise independent. The events are not

mutually independent, however, because $\Pr \{A_1 \cap A_2 \cap A_3\} = 0$ and $\Pr \{A_1\} \Pr \{A_2\} \Pr \{A_3\} = 1/8 \neq 0$.

Bayes's theorem

From the definition (C.16) of conditional probability and the commutative law $A \cap B = B \cap A$, it follows that for two events A and B , each with nonzero probability,

$$\begin{aligned} \Pr \{A \cap B\} &= \Pr \{B\} \Pr \{A \mid B\} \\ &= \Pr \{A\} \Pr \{B \mid A\} . \end{aligned} \quad (\text{C.18})$$

Solving for $\Pr \{A \mid B\}$, we obtain

$$\Pr \{A \mid B\} = \frac{\Pr \{A\} \Pr \{B \mid A\}}{\Pr \{B\}} , \quad (\text{C.19})$$

which is known as *Bayes's theorem*. The denominator $\Pr \{B\}$ is a normalizing constant, which we can reformulate as follows. Since $B = (B \cap A) \cup (B \cap \bar{A})$, and since $B \cap A$ and $B \cap \bar{A}$ are mutually exclusive events,

$$\begin{aligned} \Pr \{B\} &= \Pr \{B \cap A\} + \Pr \{B \cap \bar{A}\} \\ &= \Pr \{A\} \Pr \{B \mid A\} + \Pr \{\bar{A}\} \Pr \{B \mid \bar{A}\} . \end{aligned}$$

Substituting into equation (C.19) produces an equivalent form of Bayes's theorem:

$$\Pr \{A \mid B\} = \frac{\Pr \{A\} \Pr \{B \mid A\}}{\Pr \{A\} \Pr \{B \mid A\} + \Pr \{\bar{A}\} \Pr \{B \mid \bar{A}\}} . \quad (\text{C.20})$$

Bayes's theorem can simplify the computing of conditional probabilities. For example, suppose that you have a fair coin and a biased coin that always comes up heads. Run an experiment consisting of three independent events: choose one of the two coins at random, flip that coin once, and then flip it again. Suppose that the coin you have chosen comes up heads both times. What is the probability that it's the biased coin?

Bayes's theorem solves this problem. Let A be the event that you choose the biased coin, and let B be the event that the chosen coin

comes up heads both times. We wish to determine $\Pr \{A \mid B\}$, knowing that $\Pr \{A\} = 1/2$, $\Pr \{B \mid A\} = 1$, $\Pr \{\bar{A}\} = 1/2$, and $\Pr \{B \mid \bar{A}\} = 1/4$. Thus we have

$$\begin{aligned}\Pr\{A \mid B\} &= \frac{(1/2) \cdot 1}{(1/2) \cdot 1 + (1/2) \cdot (1/4)} \\ &= 4/5.\end{aligned}$$

Exercises

C.2-1

Professor Rosencrantz flips a fair coin twice. Professor Guildenstern flips a fair coin once. What is the probability that Professor Rosencrantz obtains strictly more heads than Professor Guildenstern?

C.2-2

Prove *Boole's inequality*: For any finite or countably infinite sequence of events $A_1, A_2, \dots$,

$$\Pr\{A_1 \cup A_2 \cup \dots\} \leq \Pr\{A_1\} + \Pr\{A_2\} + \dots. \quad (\text{C.21})$$

C.2-3

You shuffle a deck of 10 cards, each bearing a distinct number from 1 to 10, in order to mix the cards thoroughly. You then remove three cards, one at a time, from the deck. What is the probability that the three cards you select are in sorted (increasing) order?

C.2-4

Prove that

$$\Pr \{A \mid B\} + \Pr \{\bar{A} \mid B\} = 1.$$

C.2-5

Prove that for any collection of events $A_1, A_2, \dots, A_n$,

$$\begin{aligned}\Pr\{A_1 \cap A_2 \cap \dots \cap A_n\} &= \Pr\{A_1\} \cdot \Pr\{A_2 \mid A_1\} \cdot \Pr\{A_3 \mid A_1 \cap A_2\} \cdots \\ &\quad \Pr\{A_n \mid A_1 \cap A_2 \cap \dots \cap A_{n-1}\}.\end{aligned} \quad (\text{C.22})$$

★ C.2-6

Show how to construct a set of n events that are pairwise independent but such that no subset of $k > 2$ of them is mutually independent.

★ C.2-7

Two events A and B are *conditionally independent*, given C , if

$$\Pr \{A \cap B \mid C\} = \Pr \{A \mid C\} \cdot \Pr \{B \mid C\}.$$

Give a simple but nontrivial example of two events that are not independent but are conditionally independent given a third event.

★ C.2-8

Professor Gore teaches a music class on rhythm in which three students—Jeff, Tim, and Carmine—are in danger of failing. Professor Gore tells the three that one of them will pass the course and the other two will fail. Carmine asks Professor Gore privately which of Jeff and Tim will fail, arguing that since he already knows at least one of them will fail, the professor won't be revealing any information about Carmine's outcome. In a breach of privacy law, Professor Gore tells Carmine that Jeff will fail. Carmine feels somewhat relieved now, figuring that either he or Tim will pass, so that his probability of passing is now $1/2$. Is Carmine correct, or is his chance of passing still $1/3$? Explain.

C.3 Discrete random variables

A *(discrete) random variable* X is a function from a finite or countably infinite sample space S to the real numbers. It associates a real number with each possible outcome of an experiment, which allows us to work with the probability distribution induced on the resulting set of numbers. Random variables can also be defined for uncountably infinite sample spaces, but they raise technical issues that are unnecessary to address for our purposes. Therefore we'll assume that random variables are discrete.